

Low scaling BSE implementation in the exciting code

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Summary

Solving the Bethe-Salpeter Equation (BSE) is essential for understanding excited-state systems, but often challenging to converge or even computationally prohibitive for large systems. We implement a matrix-free BSE solver leveraging Interpolative Separable Density Fitting (ISDF) to interpolate electron-hole interaction kernels together with the Lanczos algorithm for diagonalization, avoiding full matrix setup. The scaling of our implementation is bounded by $\mathcal{O}(N_o N_u N_k \log N_k)$ and is thus a massive improvement over methods that set up the whole matrix, scaling with at least $\mathcal{O}((N_o N_u N_k)^3)$, where N_o and N_u are the numbers of occupied and unoccupied states, respectively, and N_k is the number k -points.

Theoretical background

The Bethe-Salpeter Equation (BSE) within many body perturbation theory (MBPT) provides the state-of-the-art frame work for describing light-matter interaction. In particular, it is used to obtain optical absorption spectra, including the effects of excitons, which are bound electron-hole states. By expanding the electron-hole wavefunctions in the transition basis, solving the BSE can be reduced to a Schrödinger like equation. Setting up and diagonalizing the Bethe-Salpeter Hamiltonian (BSH) are the computationally expensive tasks (Vorwerk et al., 2019). The BSH is given as

$$H^{BSH} = D + \gamma V - W,$$

where $\gamma = 2$ gives the spin-singlet and $\gamma = 0$ spin-triplet channel, respectively. The diagonal term D is given by the differences of the one-particle energies of the occupied (o) and unoccupied (u) states:

$$D_{ouk,o'u'k'} = (\varepsilon_{uk} - \varepsilon_{ok}) \delta_{oo'} \delta_{uu'} \delta_{kk'}.$$

The matrix elements of the repulsive exchange interaction V and the attractive screened Coulomb interaction W are calculated by solving integrals of the form

$$V_{ouk,o'u'k'} = \int d^3r \int d^3r' \frac{u_{ok}(\mathbf{r}) u_{uk}^*(\mathbf{r}) u_{o'k'}^*(\mathbf{r}') u_{u'k'}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|},$$

$$W_{ouk,o'u'k'} = \int d^3r \int d^3r' u_{uk}^*(\mathbf{r}) u_{u'k'}(\mathbf{r}) W_{k-k'}(\mathbf{r}, \mathbf{r}') u_{ok}(\mathbf{r}') u_{o'k'}^*(\mathbf{r}'),$$

where $W_{k-k'}(\mathbf{r}, \mathbf{r}')$ is the statically screened Coulomb potential. $u_{ik}(\mathbf{r})$ is the periodic part of the one-particle wavefunction of state i at the reciprocal lattice point $\mathbf{k}$. In general, we cannot see, which of the matrix elements will be zero, thus we need to compute all of them. Therefore, setting up the full BSH scales with $\mathcal{O}(N_o^2 N_u^2 N_k^2)$ and diagonalizing it directly with $\mathcal{O}(N_o^3 N_u^3 N_k^3)$.

33 The primary scaling bottleneck stems from evaluating the matrix elements of the interaction
 34 kernels. To mitigate this, we reformulate the wavefunction products using Interpolative
 35 Separable Density Fitting (ISDF) Lu & Ying (2016). Specifically, we approximate these
 36 products on a discrete real-space grid, $\{\mathbf{r}\}$, by expressing them as superpositions of values
 37 evaluated on a smaller interpolation grid, $\{\mathbf{r}_\mu\} \subset \{\mathbf{r}\}$:

$$u_{ik}^*(\mathbf{r})u_{jk'}(\mathbf{r}) \approx \sum_{\mu=1}^{N_\mu} \zeta_\mu(\mathbf{r})u_{ik}^*(\mathbf{r}_\mu)u_{jk'}(\mathbf{r}_\mu),$$

38 where N_μ is the number of interpolation points, and $\zeta_\mu(\mathbf{r})$ are the expansion coefficients.
 39 Due to the tensor product structure of $u_{ik}^*(\mathbf{r})u_{jk'}(\mathbf{r})$, $\zeta_\mu(\mathbf{r})$ can be computed efficiently, and
 40 the scaling is bounded by $\mathcal{O}(N_\mu^3)$ (Hu et al., 2017). We observe that we can always choose
 41 $N_\mu \ll N_o N_u N_k$, thus ISDF is never a bottleneck. The interpolation points are computed
 42 efficiently with centroidal Voronoi tessellation within $\mathcal{O}(N_\mu N_r)$ (Dong et al., 2018).

43 Inserting ISDF in the equations for the interaction kernels yields

$$V_{ouk,o'u'k'} \approx \frac{1}{N_k^2} \sum_{\mu=1}^{N_\mu^V} \sum_{\nu=1}^{N_\mu^V} u_{ok}(\mathbf{r}_\mu^V)u_{uk}^*(\mathbf{r}_\mu^V)\tilde{V}_{\mu\nu}u_{o'k'}^*(\mathbf{r}_\nu^V)u_{u'k'}(\mathbf{r}_\nu^V),$$

$$W_{ouk,o'u'k'} \approx \frac{1}{N_k^2} \sum_{\mu=1}^{N_\mu^{W_u}} \sum_{\nu=1}^{N_\mu^{W_o}} u_{uk}^*(\mathbf{r}_\mu^{W_u})u_{u'k'}(\mathbf{r}_\mu^{W_u})\tilde{W}_{\mu\nu,k-k'}u_{ok}(\mathbf{r}_\nu^{W_o})u_{o'k'}^*(\mathbf{r}_\nu^{W_o}),$$

45 where we have shifted the integration from the wavefunction pairs to the interpolation
 46 coefficients such that

$$\tilde{V}_{\mu\nu} = \int_{\Omega^l \times \Omega^l} d\mathbf{r}d\mathbf{r}' \zeta_\mu^* V(\mathbf{r}, \mathbf{r}') \zeta_\nu^V(\mathbf{r}'),$$

$$\tilde{W}_{\mu\nu,k-k'} = \int_{\Omega^l \times \Omega^l} d\mathbf{r}d\mathbf{r}' \zeta_\mu^* W_{k-k'}(\mathbf{r}, \mathbf{r}') \zeta_\nu^{W_o}(\mathbf{r}').$$

48 Note that there are three different wavefunction pairings, i.e. $u_{ok}^*(\mathbf{r})u_{uk}(\mathbf{r})$ for the exchange
 49 kernel (V) and $u_{ok}^*(\mathbf{r})u_{o'k'}(\mathbf{r})$, $u_{uk}^*(\mathbf{r})u_{u'k'}(\mathbf{r})$ for the screened kernel W. Each pairing
 50 requires a separate ISDF calculation, denoted by the superscripts V, W_o , and W_u . Since
 51 the number of combinations may vary, the number of interpolation points required may also
 52 vary. Reformulating the matrix elements in this way alone does not improve scaling with
 53 respect to the system size. To achieve this, we combine it with an iterative solver, here
 54 with the Lanczos algorithm. This class of algorithms constructs an approximation to the
 55 eigenvalues and eigenvectors by iteratively applying matrix-vector multiplications. Applying
 56 the interaction kernels in their interpolated forms to a vector X of dimension $N_o N_u N_k$ allows
 57 efficient computation by rearranging the summations to exploit a separable structure of the
 58 kernels. For the exchange kernel we get

$$[V \cdot X]_{ouk} = \frac{1}{N_k} \sum_{\mu=1}^{N_\mu^V} u_{uk}^*(\mathbf{r}_\mu^V)u_{ok}(\mathbf{r}_\mu^V) \left\{ \sum_{\nu=1}^{N_\mu^V} \tilde{V}_{\mu\nu} \left[\sum_{k'} \left(\sum_{u'} u_{u'k'}(\mathbf{r}_\nu^V) \left[\sum_{o'} u_{o'k'}^*(\mathbf{r}_\nu^V) \cdot X_{o'u'k'} \right] \right) \right] \right\},$$

59 where we first compute the sums over o' , u' , and k' to get a term that depends only on $\mathbf{r}_\nu^V$
 60 with a complexity of $\mathcal{O}(N_\mu^V(N_o N_u N_k + N_u N_k))$. The remaining sums can be computed with
 61 $\mathcal{O}((N_\mu^V)^2 N_\mu^V N_o N_u N_k)$, so the complexity of computing $V \cdot X$ is bounded by $\mathcal{O}((N_\mu^V)^2 +$
 62 $N_\mu^V N_o N_u N_k)$. Applying the screened kernel to X , after reordering the sums, we get

$$[W \cdot X]_{ouk} = \frac{1}{N_k} \sum_{\nu=1}^{N_\mu^{W_o}} u_{ok}(\mathbf{r}_\nu^{W_o}) \left\{ \sum_{\mu=1}^{N_\mu^{W_u}} u_{uk}^*(\mathbf{r}_\mu^{W_u}) \sum_{k'} \left[\tilde{W}_{\mu\nu,k-k'} \right. \right.$$

63

$$\times \left(\sum_{u'} u_{u'k'}(\mathbf{r}_{\mu}^{W_u}) \left[\sum_{o'} u_{o'k'}^*(\mathbf{r}_{\nu}^{W_o}) X_{o'u'k'} \right] \right) \Bigg\}.$$

64 Here we exploit the separable structure of the decomposition so that the terms depending on
65 $\mathbf{k}$ and $\mathbf{k}'$ are on the left and right of $\tilde{W}_{\mu\nu, \mathbf{k}-\mathbf{k}'}$. The evaluation of the two innermost sums
66 over o' and u' to $A_{\mu\nu}^{k'}$ scales with $\mathcal{O}(N_{\mu}^{W_u} N_o N_u N_k + N_{\mu}^{W_o} N_{\mu}^{W_u} N_u N_k)$. Then the sum over
67 $\mathbf{k}'$ reads as a discrete convolution

$$\sum_{\mathbf{k}'} W_{\mathbf{k}-\mathbf{k}'} A_{\mu\nu}^{k'},$$

68 which can be efficiently evaluated with fast Fourier transforms simultaneously within
69 the $\mathcal{O}(Nk \log Nk)$ scaling for each $\mu\nu$ pair. The remaining summations scale with
70 $\mathcal{O}(N_{\mu}^{W_o} N_{\mu}^{W_u} N_u N_k)$. So the complexity for the computation of $W \cdot X$ is bounded by
71 $\mathcal{O}(N_{\mu}^{W_u} N_o N_u N_k + N_{\mu}^{W_o} N_{\mu}^{W_u} N_u N_k + N_{\mu}^{W_o} N_{\mu}^{W_u} N_k \log N_k)$.

72 Statement of need

73 Due to the unfavorable scaling of solving the BSE directly, many interesting problems such as
74 complex materials with large unit cells or systems requiring a dense Brioullin-zone sampling
75 are not feasible. Even though Henneke and coworkers (Henneke et al., 2020) have already
76 described the new algorithm and demonstrated the scaling improvement, an easy-to-use and
77 scalable implementation was still missing. We have implemented and fully integrated this
78 approach in the existing BSE infrastructure of the all-electron, full-potential package exciting
79 (Gulans et al., 2014). Users can now easily choose which algorithm they prefer to use and
80 have the full suite of exciton analysis implemented in exciting at hand.

81 Results

82 For computing the ISDF, two new parameters, n_r and c_{μ} , are introduced. n_r is the real-space
83 sampling density for $u_{ik}(\mathbf{r})$ and is defined as

$$n_r = \frac{N_r}{\Omega},$$

84 where N_r is the number of $\mathbf{r}$ -points and Ω the unit cell volume. The sampling is chosen to be
85 regular such that the distance between the sampling points in each lattice direction is as similar
86 as possible. The dimensionless parameter c_{μ} is used to control the number of interpolation
87 points and is defined as

$$N_{\mu} = c_{\mu} \sqrt{\sqrt{N_{\text{pairs}}}},$$

88 where N_{pairs} refers to the number of wave function pairs for which ISDF is computed. Note
89 that N_{pairs} depends on N_k . The double square-root dependence ensures that the overall scaling
90 remains below $\mathcal{O}(N_k^2)$. In Fig.(Figure 1) we present, for the example of diamond, the difference
91 in exciton binding energies obtained with the new implementation and a reference calculation.
92 The reference, based on the direct implementation, sets up and diagonalizes the full BSH and
93 depends neither on n_r nor c_{μ} . The results are shown as functions of n_r and c_{μ} for a small
94 $\mathbf{k}$ -grid of $2 \times 2 \times 2$. Additionally, we show the spectral similarities compared to the reference
95 calculation as functions of n_r and c_{μ} . For both parameters, both properties converge as their
96 values increase. To find the optimal interpolation grid for ISDF, we first converge n_r , then
97 c_{μ} . In our example, $n_r = 138$ [a.u.] and $c_{\mu} = 40.0$ yield converged results. This corresponds
98 to a real-space sampling of $22 \times 22 \times 22$ and numbers of interpolation points $N_{\mu}^V = 202$,
99 $N_{\mu}^{W_o} = 322$, and $N_{\mu}^{W_u} = 360$.

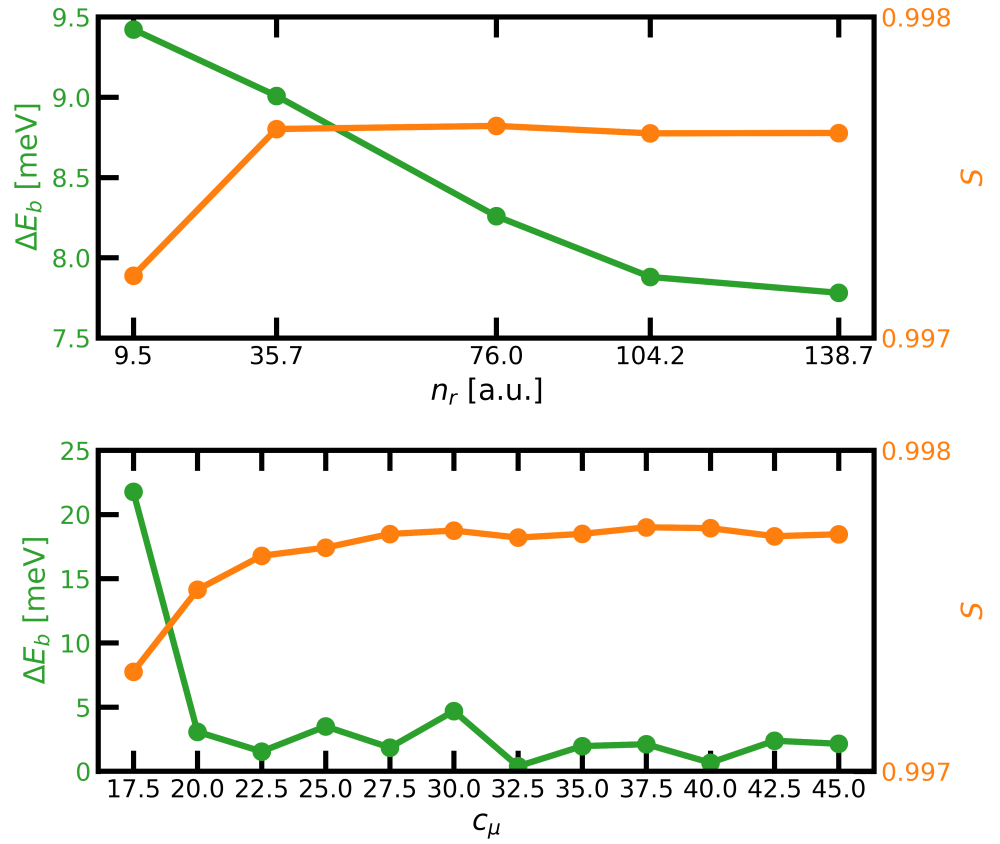


Figure 1: Difference of the exciton binding energy and spectral similarity between the new method and the direct method as functions of n_r (upper panel) and c_μ (lower panel) for diamond on a $2 \times 2 \times 2$ k-grid.

In Fig. (Figure 2) we compare the exciton binding energies and spectra of the new implementation to those of the old implementation for increasing k-grids, while keeping n_r as well as N_μ^V , $N_\mu^{W_o}$, and $N_\mu^{W_u}$ fixed at the values above. We observe that the results converge as N_k increases. Thus, the number of interpolation points is asymptotically independent of N_k . A similar behavior of ISDF was observed in (Lu & Ying, 2016).

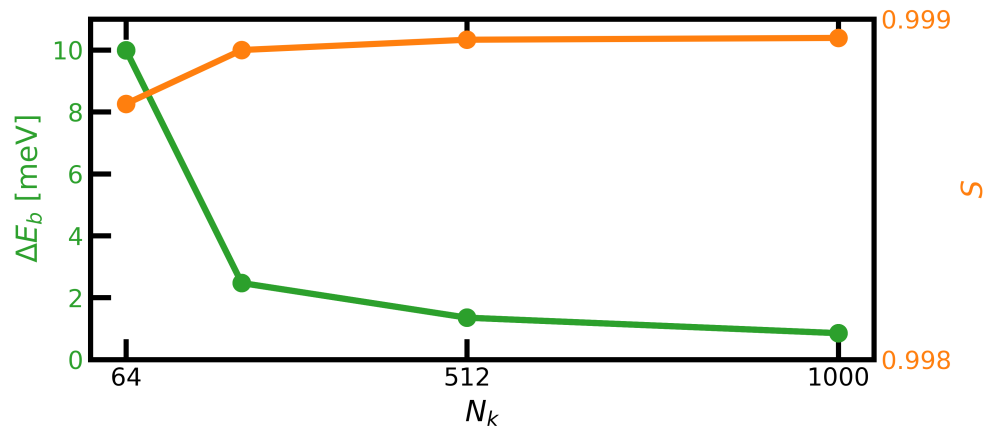


Figure 2: Difference of the exciton binding energy and spectral similarity between the new method and the direct method as functions of N_k for diamond.

In Fig.(Figure 3) we show the wall times for solving the BSE with the new and direct implementations for increasing N_k . The new algorithm massively outperforms the direct one, and the speedup increases more than linearly with N_k . We also show the wall times for computing the RPA screening with increasing N_k , which is now clearly the bottleneck in solving the BSE.

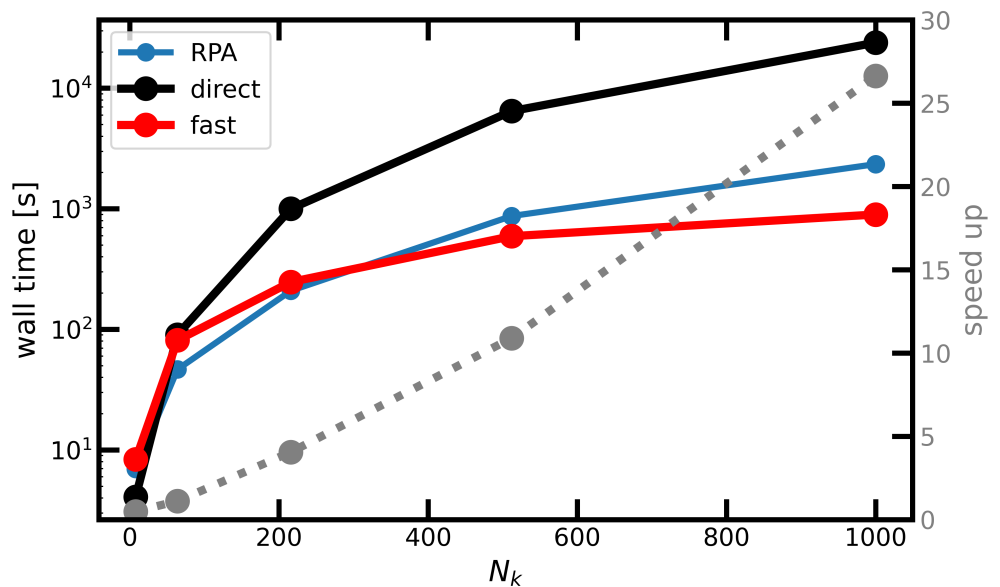


Figure 3: Runtimes of the direct (black) and new (red) BSE implementations and the RPA screening (blue) as functions of N_k . The speedup of the algorithm is shown by the gray dashed line.

Altogether, we have implemented a new, low-scaling BSE solver and fully integrated it into the all-electron, full-potential package exciting. We demonstrate that the new implementation yields results equivalent to the direct solution of the BSE but with significantly reduced computational time. Consequently, it enables more precise calculations and facilitates the study of more complex problems.

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